

Exp 12:

Write the python program to implement Minimax algorithm for gaming.

Aim

To write and implement a Python program for the Minimax algorithm, which helps a player choose the best possible move in a two-player game by maximizing its chances of winning while minimizing the opponent's chances.

Algorithm

1. Start the program.
2. Represent the game state (board or game tree).
3. Check whether the current state is a terminal state (win, lose, or draw).
4. If it is a terminal state, return the evaluation score.
5. If it is the maximizing player's turn:
6. Generate all possible moves.
7. Apply the Minimax function recursively for each move.
8. Choose the move with the highest score.
9. If it is the minimizing player's turn:
10. Generate all possible moves.
11. Apply the Minimax function recursively for each move.
12. Choose the move with the lowest score.
13. Repeat the process until all possible moves are evaluated.
14. Return the best move to the current player.
15. Display the optimal move and its score.
16. End the program.

Program:

Minimax Algorithm

```
def minimax(depth, nodeIndex, isMax, values, height):
```

```
    # Base case: reached leaf node
```

```
    if depth == height:
```

```
        return values[nodeIndex]
```

```
    if isMax:
```

```

    return max(
        minimax(depth + 1, nodeIndex * 2, False, values, height),
        minimax(depth + 1, nodeIndex * 2 + 1, False, values, height)
    )
else:
    return min(
        minimax(depth + 1, nodeIndex * 2, True, values, height),
        minimax(depth + 1, nodeIndex * 2 + 1, True, values, height)
    )

# Leaf node values
values = [3, 5, 2, 9, 12, 5, 23, 23]

# Height of the game tree
height = 3

# Find the optimal value
result = minimax(0, 0, True, values, height)

print("Optimal Value:", result)

```

Output:

```

---
>>>
Optimal Value: 12
>>> |

```

Result:

The Python program implementing the Minimax algorithm was successfully executed. The algorithm evaluated all possible game moves recursively and selected the optimal move, ensuring the best possible outcome for the current player.